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MANAGEMENT INFORMATION SYSTEMS**



FINAL ESSAY

TOPIC:

**ANALYSIS AND DESIGN OF THE CK GO E-COMMERCE
AND LOYALTY SYSTEM**

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INTRODUCTION: OVERVIEW OF THE CHOSEN BUSINESS

Circle K is a globally recognized convenience store chain that was founded in 1951 by entrepreneur Fred Hervey in El Paso, Texas, etc. Today, it operates as a flagship subsidiary of the Canadian multinational company Alimentation Couche-Tard, Inc. With over 14,800 stores worldwide, Circle K is famous for providing a wide array of convenience products, fresh food, and excellent customer service. In Vietnam, Circle K has become a dominant player in the 24/7 convenience retail market.

Despite its massive physical footprint, recent market analyses have highlighted that Circle K historically possessed a limited e-commerce presence compared to digital-first competitors. Recognizing the rapid shift in consumer behavior and the necessity of digital adaptation, Circle K Vietnam has launched "CK GO".



Picture 1. Circle K and CK GO official logo

The Official Logo of Circle K Note. Copyright by Circle K IP Management, LLC.

CK GO is a comprehensive mobile e-commerce and loyalty application designed to bridge the gap between offline and online retail. The system allows customers to create accounts, place online orders for home delivery, and utilize secure cashless payment methods via banks or third-party gateways. Furthermore, CK GO integrates a robust loyalty program, it tracks purchase history to accumulate points, eventually upgrading customers to exclusive membership tiers, such as Silver and Gold, which unlock special promotional benefits.

This essay will focus specifically on the CK GO E-commerce and Loyalty Management System. There are two primary reasons for selecting this specific system:

1. **Strategic Business Value:** Addressing the limited e-commerce presence is a critical step for Circle K. Leveraging e-commerce for home deliveries has been identified as a major growth opportunity that can significantly boost the company's revenue and customer reach.
2. **System Complexity:** From a systems analysis perspective, the CK GO application presents a dynamic environment with complex business rules. For instance, calculating delivery fees and discounts involves intricate decision-making based on the customer's membership tier (Silver/Gold), cart subtotal, and payment method (cashless versus cash on delivery). This provides an excellent real-world scenario for in-depth system modeling.

PART 1: ANALYZING THE ORGANIZATIONAL SYSTEM AND DEFINING SYSTEM BOUNDARIES USING DFD

1. Analyzing the organizational system and defining system boundaries using DFD

1.1 Systems Perspective Analysis of the Organization

According to Kendall's book, organizations and their members are usefully conceptualized as complex systems designed to accomplish predetermined goals. All systems process inputs from their environments and transform them into outputs. To thoroughly understand the CK GO e-commerce and loyalty platform, it is crucial to analyze its fundamental operations through the lens of four core system components: Input, Processing, Output, and Response.

1. **Input:** An organization must be able to import resources and information through its boundaries to function. For the CK GO system, inputs consist of the massive streams of data actively and passively generated by users. Based on Circle K Vietnam's data policies, inputs include customer identification details (name, phone number, email, and exact delivery address), mobile device GPS locations, and specific order data (item selections and quantities). Additionally, financial data, such as credit card or e-wallet details, serve as critical inputs for cashless transactions.
2. **Processing:** Processes are the actions that transform inputs into outputs, involving activities like verifying, calculating, and updating (Kendall & Kendall, 2020, p. 30). Within the CK GO system, several activities occur simultaneously: the system verifies user credentials, evaluates purchase history to determine Loyalty Tiers (Standard, Silver, or Gold), and calculates totals.

- Crucially, the system performs Spatial Processing by determining delivery fees based on GPS coordinates and routing requests to the nearest store location.
3. **Output:** Outputs are the finished products, services, or information that the system exchanges with the outside world. The transformed data exits the CK GO system in various formats tailored to different entities. For the customer, the outputs are electronic order confirmations, digital receipts, updated loyalty point balances, and ultimately, the physical delivery of the purchased goods. For the internal store operations, the output is a generated "picking list" that guides store clerks in gathering the requested items from the shelves.
 4. **Response (Feedback):** Feedback is a critical form of system control; it uses system outputs to compare actual performance against organizational goals, helping managers formulate more specific strategies. In the CK GO ecosystem, feedback manifests as customer ratings, delivery time metrics, out-of-stock alerts, and responses to promotional surveys. Circle K utilizes this feedback data to perform statistical analysis, assess risks, adjust product offerings, and ultimately improve the overall shopping experience. This self-regulating feedback loop ensures that the CK GO system adapts to changing consumer demands and remains highly competitive in the convenience retail market.

1.2 Identification of External Entities and System Boundaries

According to Kendall & Kendall (2020), all systems are contained by boundaries separating them from their environments. Organizational boundaries exist on a continuum ranging from extremely permeable to almost impermeable. Anything external to an organization's boundaries is considered to be its environment. To properly analyze the CK GO platform, we must clearly define what is inside the system and what remains outside.

The system boundary of the CK GO E-commerce and Loyalty System encompasses the mobile application interface, the order management processing engine, the internal product inventory database, and the loyalty points calculation logic. It is a relatively open system that allows the free passage of information through its boundaries. However, it strictly excludes the physical in-store stocking procedures, the internal banking operations, and the internal routing algorithms of third-party logistics companies.

An external entity represents any entity (a person, a group, a department, or another system) that supplies or receives information from the system but is not a part of the system. For the CK GO system, the primary external entities interacting across the system boundaries are:

1. **Customer:** The primary human entity that originates the order and supplies personal data, order details, and payment information. In return, they receive the final goods, order confirmations, and digital receipts.

2. **Payment Gateway (Bank / E-wallet):** An external financial system. CK GO sends payment requests to this entity and receives payment verification statuses in return. Circle K does not control how the bank processes the transaction internally.
3. **Delivery Partner:** An external logistics service (e.g., a third-party shipping company or independent driver). The system sends them delivery requests and customer addresses, and they return delivery status updates.

Additionally, the Customer provides feedback and ratings, which the system processes to improve service quality.

1.3 Context-Level Data Flow Diagram (DFD)

The context-level data flow diagram contains only one process, representing the entire system, and defines the boundaries of the system. It provides a bird's-eye view of data movement in the system and the broadest possible conceptualization.

The diagram below illustrates the external entities and the major data flows interacting with the central CK GO Management System.

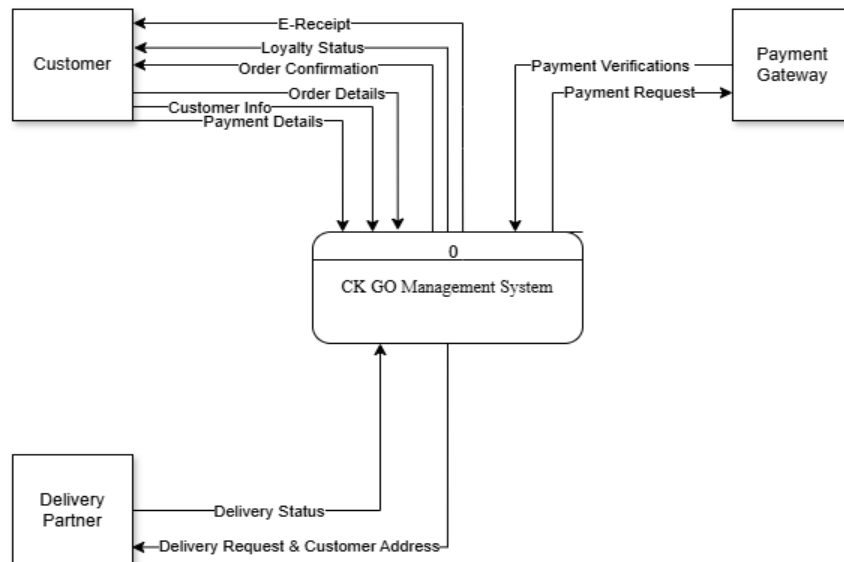


Figure 1.3-1 Context-Level Data Flow Diagram for the CK GO Management System

(Source: Created by the author based on Kendall & Kendall, 2020)

1.4 Level 0 Data Flow Diagram

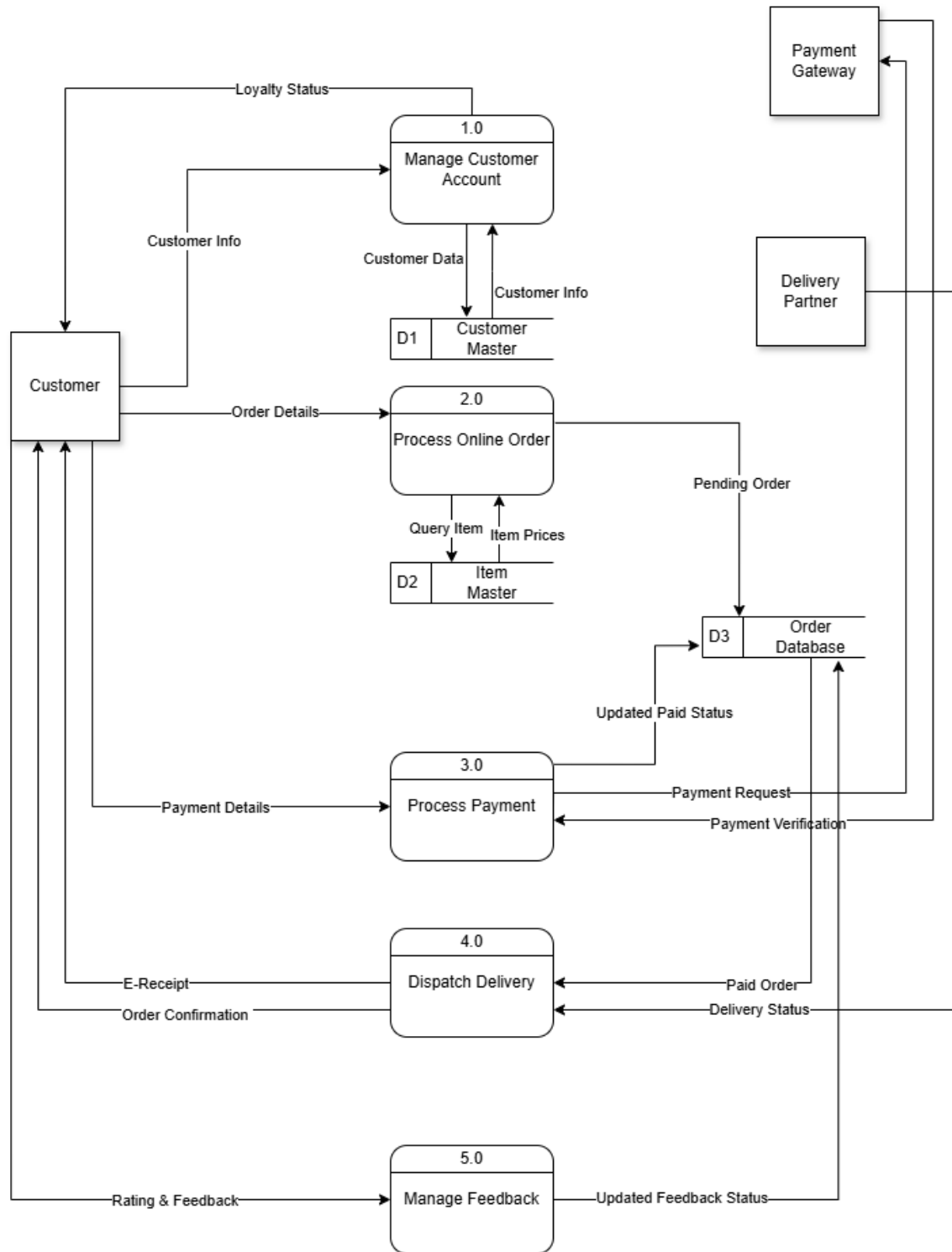


Figure 1.4-1 Level 0 Data Flow Diagram for the CK GO Management System
(Source: Created by the author based on Kendall & Kendall, 2020)

According to Kendall & Kendall (2020), Diagram 0 expands on the context diagram by introducing data stores and identifying the major business processes without overcomplicating the model. The CK GO Level 0 DFD consists of three internal data stores and four primary processes that interact to fulfill customer requests:

1. Data Stores Identified:

- 1) **D1 Customer Master:** A permanent master file that stores customer demographic information, account credentials, and accumulated loyalty points (Silver/Gold tiers).
- 2) **D2 Item Master:** A database storing the product catalog, including current item prices, descriptions, and inventory availability.
- 3) **D3 Order Database:** A transaction file that stores all customer orders, capturing the order status as it moves from pending to paid, and eventually to dispatched.

2. Detailed Process Logic:

- 1) **Process 1.0 (Manage Customer Account):** This process is responsible for handling customer registration and profile updates. It receives Customer Info from the Customer entity, updates and retrieves the Customer Profile from the D1 Customer Master data store, and outputs the calculated Loyalty Status back to the user interface.
- 2) **Process 2.0 (Process Online Order):** Upon receiving Order Details from the Customer, this process sends a Query Item request to the D2 Item Master to retrieve valid Item Prices. It then calculates the cart subtotal and generates a Pending Order, which is securely saved into the D3 Order Database.
- 3) **Process 3.0 (Process Payment):** This financial process retrieves the Pending Order from D3 and takes the Payment Details input from the Customer. It sends a Payment Request to the external Payment Gateway. Once the Payment Verification is successfully returned, this process saves the Updated Paid Status back to the D3 Order Database.
- 4) **Process 4.0 (Dispatch Delivery):** The final operational process reads the completed Paid Order from D3. It extracts the customer's address and sends a Delivery Request to the Delivery Partner. As the order moves, it receives Delivery Status updates from the partner and ultimately issues an E-Receipt and Order Confirmation to the Customer.

1.5 Interaction of Different Management Levels with Data (Strategic, Managerial, Operational)

According to Kendall & Kendall (2020, p.32), management in organizations exists on three broad, horizontal levels: operational control, managerial planning and control, and strategic management. Each level carries its own responsibilities and interacts differently with the data generated by the CK GO system:

- **Operational Control:** At this level, operations managers make decisions based on predetermined rules with predictable outcomes. At Circle K, store clerks and shift supervisors utilize transactional, real-time data from the CK GO application. They rely on specific Order Details and Order Picking Slips to perform repetitive, high-precision tasks, such as accurately gathering items for handover to the Delivery Partner. Information at this level is internal, low-level, and highly structured.
- **Managerial Planning and Control (Middle Management):** Middle managers focus on short-term planning and resource allocation. Regional managers at Circle K require summarized, tactical information in real-time to assess performance across locations. By analyzing daily and weekly metrics—such as regional order volumes, delivery latency, and inventory shortages—they make critical decisions regarding inventory reordering schedules and personnel allocation to mitigate traffic spikes in high-demand zones.
- **Strategic Management:** Strategic managers adopt a long-term, outward-looking perspective to guide the organization's future. Top executives at Circle K Vietnam utilize aggregated, analytical data to create "what-if" scenarios and predictive models. They analyze long-term trends, such as the growth rate of high-tier loyalty members (Gold/Silver) and shifts in cashless payment adoption, to justify large-scale investments in e-commerce infrastructure, nationwide marketing strategies, or the establishment of new distribution centers.



Figure 1.5-1 Management Levels and Information Requirements.

(Created by author based on Kendall and Kendall 2020)

PART 2: BUILDING A COMPREHENSIVE INFORMATION REQUIREMENTS GATHERING STRATEGY

2. Building a comprehensive information requirements gathering strategy

2.1. Interview Plan Design

An information-gathering interview is a directed conversation with a specific purpose that uses a question-and-answer format to elicit opinions, feelings, and goals from organization members

2.1.1 Identification of Interview Subjects

To ensure a balanced perspective that addresses different management levels, two key subjects have been selected for personal interviews:

2.1.2 Selection of Interview Structure

For the interview with the E-commerce Operations Manager, a Funnel Structure will be utilized. According to Kendall & Kendall (2020), the funnel structure takes a deductive approach by beginning with generalized, open-ended questions and then narrowing the possible responses by using closed questions.

This structure is highly appropriate for interviewing a manager because it provides an easy, nonthreatening way to begin the interview. It allows the manager to freely express their broad vision and emotional feelings regarding the shift to e-commerce

before the analyst narrows the focus to gather specific, quantifiable data about the CK GO system's technical rules

2.1.3 Sample Interview Questionnaire

The following is the planned sequence of questions for the E-commerce Operations Manager, demonstrating the funnel approach from broad opinions to specific facts:

- **Question 1 (Open-ended / Broad):** "What is your overall vision for the CK GO application in the Vietnamese convenience retail market over the next two years?"
- **Question 2 (Open-ended):** "What are the biggest frustrations your team has experienced during the transition from traditional offline sales to handling online home deliveries?"
- **Question 3 (Probe):** "You mentioned 'delivery fee miscalculations' as a major frustration. Could you give a specific example of how this issue impacts customer satisfaction on the app?"
- **Question 4 (Closed / Specific):** "Currently, are the delivery fees on the CK GO app calculated based on both the geographic distance and the customer's VIP membership tier?"
- **Question 5 (Closed / Very Specific):** "What is the exact minimum cart subtotal required for a Gold-tier member to receive a 10% discount and free shipping on their online order?"

2.2. Broad Survey Questionnaire for End Users

Questionnaires are an interactive information-gathering technique that allows systems analysts to study attitudes, beliefs, behavior, and characteristics of several key people who may be affected by the current and proposed systems. Because the target audience for the CK GO system is a broad, widely dispersed sample of end users (mobile shoppers), a self-administered Web survey is the most efficient administration method.

To accurately quantify the customers' feedback regarding the system's human-computer interaction, e-commerce usability, and overall usefulness, this questionnaire employs both Nominal scales to classify respondents and Interval scales (where the intervals between each pair of numbers are equal, allowing for mathematical analysis).

Table 2.2-1 End-User Survey Questionnaire for the CK GO Application
(Administration Method: Web-based survey distributed via email link to registered members))

PART A: USER CHARACTERISTICS (NOMINAL SCALE)						
1. How long have you been using the CK GO application?						
☐ Less than 1 month ☐ 1 - 6 months ☐ 6 - 12 months ☐ More than 1 year						
2. What is your current Loyalty Membership Tier?						
☐ Standard ☐ Silver ☐ Gold						
PART B: SYSTEM USABILITY & USEFULNESS (INTERVAL SCALE)						
<i>Please indicate your level of agreement with the following statements by checking the appropriate number. (1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)</i>						
No.	Evaluation Statements	1	2	3	4	5
3	The application's interface is visually appealing and intuitive to navigate.	☐	☐	☐	☐	☐
4	The checkout process is seamless and easy to complete.	☐	☐	☐	☐	☐
5	The system calculates delivery fees and loyalty discounts accurately based on my VIP tier.	☐	☐	☐	☐	☐
6	I trust the system to keep my personal data and cashless payment details secure.	☐	☐	☐	☐	☐
7	The app provides clear, immediate feedback (e.g., order confirmation, out-of-stock alerts).	☐	☐	☐	☐	☐
PART C: OPEN-ENDED QUESTION						
8. In your opinion, what is the single most frustrating problem you experience when placing an online order on CK GO, and how could we improve it?						

2.3. Application of Unobtrusive Methods

2.3.1 Applying the STROBE Method to Observe the Actual Work Environment

To implement STROBE effectively for the CK GO system analysis, an anecdotal checklist is utilized. This matrix compares the organizational narrative (what the Store Shift Supervisor said) against the observable environmental elements (what they actually do). To properly evaluate this comparison, the analyst must apply Kendall and Kendall's

(2020) five specific shorthand symbols: "1. A check mark means the narrative is confirmed. 2. An "X" means the narrative is reversed. 3. An oval or eye-shaped symbol serves as a cue for the systems analyst to look further. 4. A square means observation of the STROBE elements modifies the narrative. 5. A circle means the narrative is supplemented by what is observed"

Table 2.3-1. STROBE Anecdotal Checklist for the Store Shift Supervisor (Based on Kendall & Kendall's 5 Shorthand Symbols)

Key (Shorthand Symbols)			
(✓) Confirms the narrative (X) Negates or reverses the narrative (O) Cue to look further (Oval) (□) Modifies the narrative (Square) (○) Supplements the narrative (Circle)			
Narrative Portrayed by the Shift Supervisor (From Interview)	Props & Equipment (Devices/PCs)	Office Lighting & Color	Clothing of the Decision Maker
We use the latest technology to process online orders instantly.	X	○	✓
The store environment is highly organized and stress-free.	□	X	O
Our team is very dynamic and ready for e-commerce.	✓	✓	✓
We always check the CK GO app on our tablets immediately.	O	○	□

The STROBE analysis reveals a gap in the organizational narrative. While the supervisor claims instant order processing, the 'X' and 'Square' symbols in the Props & Equipment category suggest that the physical devices may hinder the system's efficiency. This observation is crucial for the proposed system design, as it highlights the need for hardware upgrades alongside software implementation.

2.3.2 Investigating Hard Data and Sampling

To properly investigate the quantitative hard data for the CK GO system, the analyst must examine existing organizational documents. According to Kendall and Kendall (2020), *"many quantitative documents are available for interpretation in any business, and they include reports used for decision making, performance reports, records, and a variety of forms"*.

Because examining every single printed document is too costly and time-consuming, the systems analyst must use sampling. *"Sampling is the process of systematically selecting representative elements of a population"*. To systematically analyze the system's generated forms, the analyst follows Kendall and Kendall's *"four steps to design a good sample"*:

Step 1. Determine the data to be collected or described: The analyst will collect specific quantitative documents, namely the physical "Order Picking Slips" generated by the system for store clerks. The objective is to check for missing data, unfulfilled items, or bottlenecks in the physical preparation of online orders.

Step 2. Determine the population to be sampled: The population includes all picking slips printed at the central Circle K store during a typical one-week period.

Step 3. Choose the type of sample: Out of the four main types of samples, a **purposive sample** is utilized. As Kendall and Kendall (2020) define, *"a purposive sample is based on judgment"*. The systems analyst chooses elements according to specific criteria: picking slips printed exclusively during peak online shopping hours (11:00 AM - 1:00 PM), as this is when the system and staff are under the most operational stress.

Step 4. Decide on the sample size: To balance the time required by the systems analyst and the need for accurate representation, a sample size of 50 picking slips will be selected and analyzed.

Comparing these officially generated forms with the actual actions of the staff helps identify usability issues and discrepancies within the store's operational environment.

2.4. Comparative Analysis: The Difference Between "What They Say" and "What They Do"

According to Kendall and Kendall (2020), unobtrusive methods are insufficient information-gathering methods when used alone. Instead, they must be combined with interactive methods in a multiple methods approach. *"Using both interactive and unobtrusive methods in approaching an organization is a wise practice that will result in a more complete picture of human information requirements"*.

By comparing the results obtained from interactive methods (interviews and surveys) with unobtrusive methods (STROBE and document sampling), the systems analyst can evaluate the discrepancies between the organizational narrative (what they say) and the actual observable environment (what they do). As Kendall and Kendall (2020) state, *"It often is possible to observe the particulars of the surroundings that will*

confirm or negate the organizational narrative (also called stories or dialogue) that is found through interviews or questionnaires".

Based on the STROBE Anecdotal Checklist (Table 2) and the quantitative document sampling, several key differences between the Shift Supervisor's narrative and the actual environment were identified:

- **The "X" (Negation):** During the interview, the supervisor claimed, "We use the latest technology to process online orders instantly." However, STROBE observation of the "Props & Equipment" revealed a direct negation (X). During peak hours, staff heavily relied on printing physical "Order Picking Slips" because the store's shared tablets were frequently lagging. The observable elements clearly negate the narrative, revealing a critical need for a more streamlined, responsive digital order-picking interface (e.g., a real-time mobile dashboard) to eliminate the reliance on easily lost paper slips.
- **The "□" (Modification):** The supervisor stated, "We always check the CK GO app on our tablets immediately." However, observation of the equipment modifies this narrative (□): employees actually relied more on their personal smartphones rather than the store's shared tablets, because the tablets were stationed too far from the picking shelves. This discrepancy suggests that the proposed CK GO internal management system must be optimized for mobile screens (responsive design) rather than just tablet or desktop interfaces.

By contrasting these methods, uncovering the true operational bottlenecks and hidden human information requirements that would have been completely missed if only interactive interviews were conducted.

PART 3: BUSINESS PROCESS SPECIFICATION AND DECISION-MAKING RULES IN THE SYSTEM

3. Business Process and Decision-making rules for the CK GO System

3.1. Description of a Complex Business Situation

The CK GO application features a dynamic business rule for calculating delivery fees and applying loyalty discounts. The final calculation is based on three complex conditions:

1. **Loyalty Tier:** The customer's membership status (Gold, Silver, or Standard).
2. **Cart Subtotal:** Whether the total value of items in the cart reaches the minimum threshold of 150,000 VND.
3. **Delivery Distance:** Whether the delivery address is within the standard radius ($\leq 3\text{km}$) or falls in the extended zone ($> 3\text{km}$).

Based on these combinations, the system will apply specific actions, including granting a 10% or 5% discount on the subtotal, charging a Base Shipping Fee (15,000 VND), charging an Extra Distance Surcharge (5,000 VND), or waiving the shipping fee entirely (Free Shipping).

3.2. Processing Steps Description Using Structured English

When process logic involves formulas, iteration, or structured decisions, an appropriate technique for analyzing the decision process is the use of Structured English. It is based on structured logic and simple English statements (such as IF, THEN, ELSE, DO WHILE) combined with proper indentation to show the hierarchy of the decision process.

The following Structured English specifies the logic for the "Calculate Final Order Amount" process in the CK GO system:

Process Name: Calculate Final Order Amount **Process Description:** Determines the final checkout amount by calculating cart subtotals, applying loyalty discounts, and computing distance-based shipping fees.

Process Logic:

DO WHILE there are items in the Customer Shopping Cart

 Multiply Item Price by Quantity Ordered giving Item Total

 Add Item Total to Cart Subtotal

ENDDO

IF Customer Tier is GOLD

 THEN

 Calculate Discount = Cart Subtotal * 10%

 IF Cart Subtotal >= 150,000 VND

 THEN Set Shipping Fee = 0 (Free Shipping)

 ELSE

 IF Delivery Distance <= 3km

 THEN Set Shipping Fee = 0 (Free Shipping)

ELSE Set Shipping Fee = Extra Distance Surcharge

ENDIF

ENDIF

ELSE IF Customer Tier is SILVER

THEN

Calculate Discount = Cart Subtotal * 5%

IF Cart Subtotal $\geq$ 150,000 VND

THEN Set Shipping Fee = 0 (Free Shipping)

ELSE

IF Delivery Distance $\leq$ 3km

THEN Set Shipping Fee = Base Shipping Fee

ELSE Set Shipping Fee = Base Shipping Fee + Extra Distance Surcharge

ENDIF

ENDIF

ELSE (Customer Tier is STANDARD)

THEN

Calculate Discount = 0

IF Delivery Distance $\leq$ 3km

THEN Set Shipping Fee = Base Shipping Fee

ELSE Set Shipping Fee = Base Shipping Fee + Extra Distance Surcharge

ENDIF

ENDIF

Subtract Discount from Cart Subtotal giving Discounted Subtotal Add Shipping Fee to Discounted Subtotal giving Final Order Amount.

3.3. Decision Table Construction

According to Kendall and Kendall (2020), a decision table is a table of rows and columns, separated into four quadrants: conditions, condition alternatives, actions, and rules. It is a powerful tool used to analyze structured decisions and effectively avoid impossible situations, redundancies, and contradictions.

Based on the Structured English in Section 3.2, the CK GO calculation involves three conditions: Customer Loyalty Tier, Cart Subtotal, and Delivery Distance. Initially, multiplying the alternatives for each condition creates a matrix of 12 possible rules (3 Tiers $\times$ 2 Subtotal states $\times$ 2 Distance states). However, to simplify the table, the rules can be combined where a condition alternative does not make a difference in the outcome.

For instance, if one Gold or Silver member's cart reaches 150,000 VND, they automatically receive Free Shipping regardless of the delivery distance. Therefore, the distance condition can be bypassed (indicated by a dash "-"). Similarly, Standard members do not receive free shipping regardless of their cart subtotal, so their subtotal condition is marked with a dash.

Table 3.3-1 Extended-Entry Decision Table for CK GO Order Calculation

Conditions and Actions	R1	R2	R3	R4	R5	R6	R7	R8
Conditions								
Customer Loyalty Tier	G	G	G	S	S	S	ST	ST
Cart Subtotal >= 150,000 VND	Y	N	N	Y	N	N	-	-
Delivery Distance <= 3km	-	Y	N	-	Y	N	Y	N
Actions								
A1: Apply 10% Discount	X	X	X					
A2: Apply 5% Discount				X	X	X		
A3: Apply 0% Discount							X	X
A4: Charge 0 VND (Free Shipping)	X	X		X				
A5: Charge Base Shipping (15,000 VND)					X		X	
A6: Charge Extra Surcharge (5,000 VND)			X					
A7: Charge Base + Extra (20,000 VND)						X		X

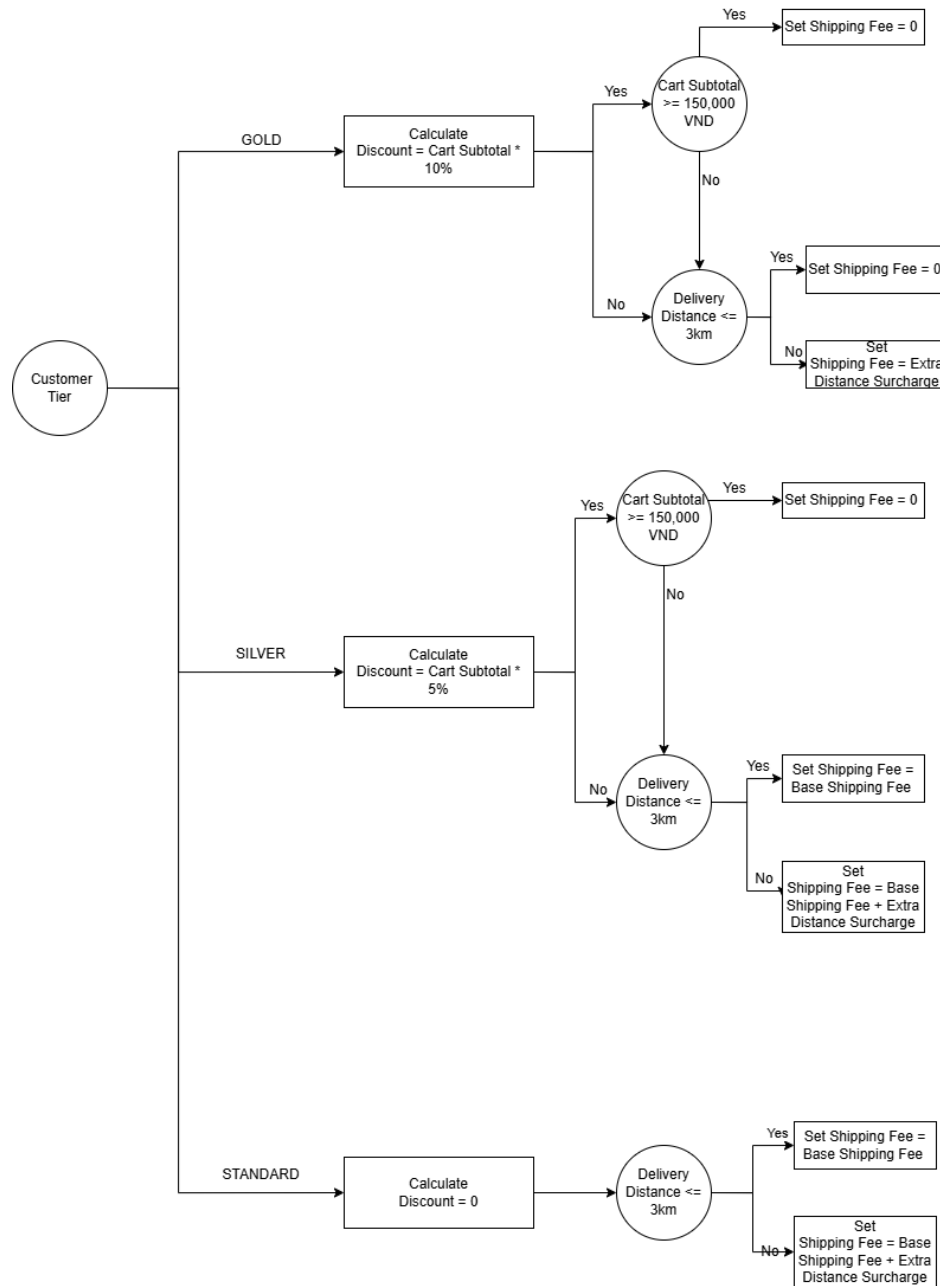
(Key: G = Gold, S = Silver, ST = Standard, Y = Yes, N = No, X = Execute Action, -) = Condition does not affect the outcome)

3.4. Decision Tree Development

According to Kendall and Kendall (2020), "decision trees are used when complex branching occurs in a structured decision process. Trees are also useful when it is essential to keep a string of decisions in a particular sequence". A decision tree has advantages over a decision table because "it takes advantage of the sequential structure of decision tree branches so that the order of checking conditions and executing actions is immediately noticeable".

In the CK GO system, the sequence of checking conditions is critical. The system first identifies the Customer Loyalty Tier (Gold, Silver, or Standard). Depending on the tier, it checks the Cart Subtotal, and finally evaluates the Delivery Distance to assign the exact shipping fee. The decision tree below directly corresponds to the 8 simplified rules established in the Decision Table (Table 3).

Table 3.4-1 Decision Tree Development based on Customer Tier



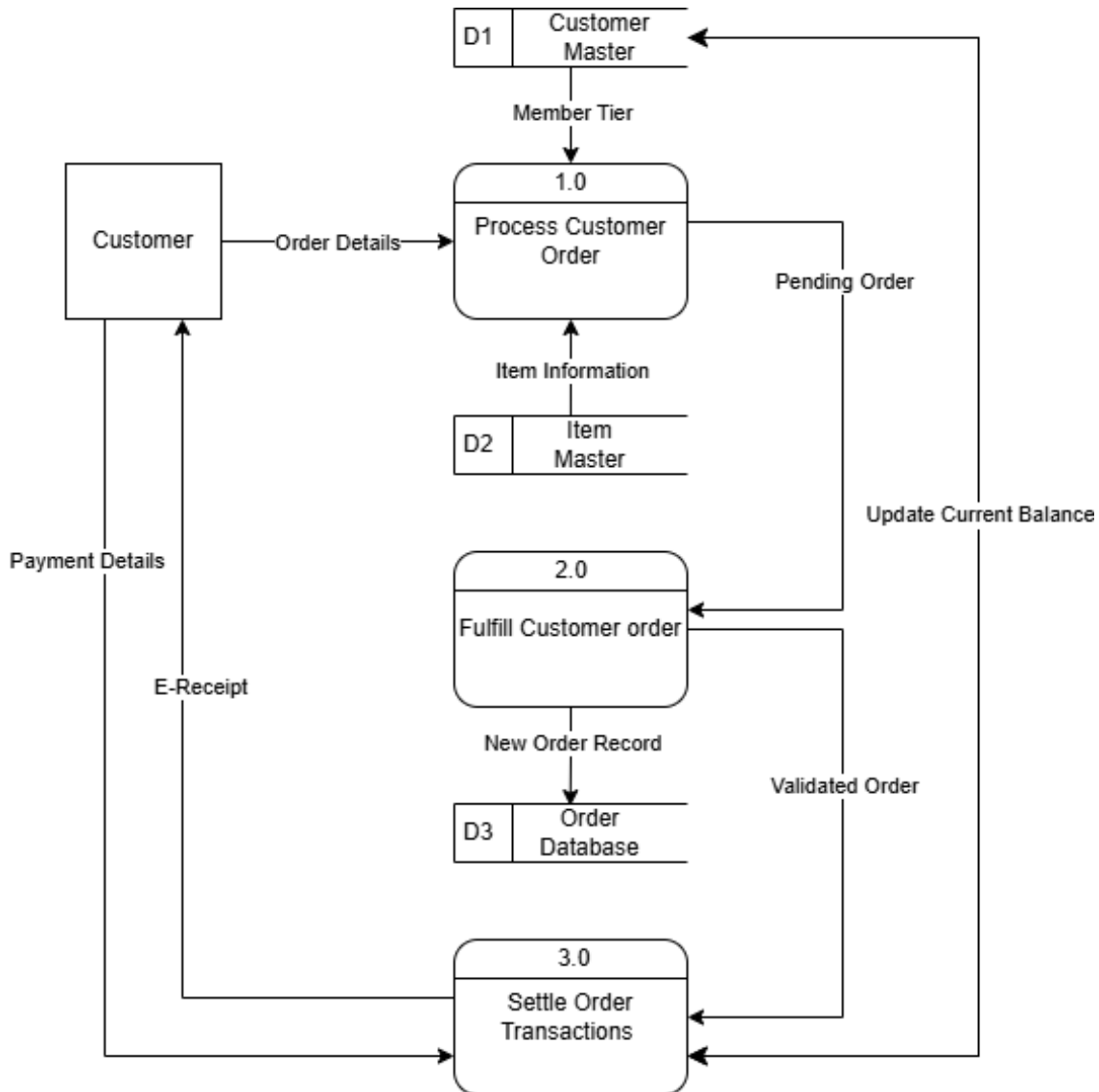
(Created by the author using Draw.io)

PART 4: DATA FLOW DIAGRAMS AND DATA DICTIONARY DESIGN FOR THE CK GO SYSTEM

4. Data Flow Diagrams and Data Dictionary Design for the CK GO System.

4.1. Level 0 DFD for the CK GO Ordering and Payment Process

4.1-1 Level 0 DFD for "Ordering and Payment process"



(Created by the author using Draw.io)

Unlike the DFD in Part 1, which depicts the general context of the entire organization, this DFD focuses exclusively on the "Ordering and Payment" process. This transition from a general organizational view to a specific logical fragment seamlessly prepares the groundwork for compiling the Data Dictionary in the following sections.

The previous name of Process 3.0 in the level 0 DFD of a entire business is Process Payment. However, by using the functional decomposition method to draw a level 0 DFD for Ordering and Payment process, was renamed as Settle Order Transactions.

4.2. Data Dictionary Entries Design

4.2.1 Data Flow Specifications

The following tables describe two major data flows within the CK GO ordering and payment sub-system: the Customer Order and the Payment Details. These entries correspond directly to the data flow arrows interacting with the external entity in the Level 0 DFD.

Table 4.2-1. Data Flow Description – Order Details (Customer Order)

DATA FLOW DESCRIPTION FORM	
Name: Order Details	
Alias: Customer Order, Online Order Request	
Description: Contains the detailed list of items selected by the customer, quantity, and specific delivery preferences initiated via the CK GO app.	
Data Flow Characteristics	
Source: External Entity: Customer	Destination: Process 1.0: Process Customer Order
Data Structure: Order Details Structure	Volume/Time: 200 orders/ day
Type of Data Flow: ☒ Screen ☐ File ☐ Report ☐ Form ☐ Internal	
Comment: This is the primary input that triggers the fulfillment process. It must be validated against the Item Master (D2).	

Table 4.2-2. Data Flow Description - Payment Details

DATA FLOW DESCRIPTION FORM	
Name: Payment Details	
Alias: Transaction Details, Payment Record	
Description: Contains the customer's specific payment method, transaction amount, and authorization data required to settle the order.	
Data Flow Characteristics	
Source: External Entity: Customer	Destination: Process 3.0: Settle Order Transactions
Data Structure: Payment Details Structure	Volume/Time: 200 payments / day
Type of Data Flow: ☒ Screen ☐ File ☐ Report ☐ Form ☐ Internal	
Comment: Payment details are sensitive data and are verified through the external Payment Gateway before an E-Receipt is generated.	

4.2.2 Data Structure Specifications

According to Kendall and Kendall (2020, p. 259), algebraic notation is used to describe data structures. Specific symbols dictate the nature of the elements: *"Braces {} indicate repetitive elements... brackets [] are used for an either/or situation... parentheses () are used for an optional element"* (p. 259).

A. Data Structure for adding order details:

Order Details = Customer_Number + Customer_Name + (Apartment) + Street_Address + Telephone + {Available_Order_Items} + Merchandise_Total + (Discount_Code) + Order_Total + Method_Of_Payment

Available_Order_Items = Item_Number + Item_Description + Quantity_Ordered + Unit_Price

Method_Of_Payment = [Credit Card | E-Wallet | Cash]

B. Data Structure for adding Payment Details:

Payment Details = Order_Number + Customer_Name + Transaction_Date + Transaction_Time + {Item_Description + Line_Item_Total} + Merchandise_Total + (Tax) + Order_Total + Payment_Method_Selection

Payment_Method_Selection = [Credit Card | E-Wallet | Cash | Banking Transfer]

4.2.3 Data Store Specifications

Data stores contain information of a permanent or semipermanent nature. The following is the data dictionary entry for the system's Customer Master file (D1), which interacts directly with the Settle Order Transaction process to update the customer's current balance and reward points.

Table 4.2-3: Data Store Description - Customer Master

DATA STORE DESCRIPTION FORM (ID: D1)	
Name: Customer Master	
Alias: Client Master, Registered User File	
Description: Contains a permanent record for each registered CK GO customer, including their profile and account status.	
Data Store Characteristics:	
File Type: ☒ Computer ☐ Manual	
File Format: ☒ Database ☐ Indexed ☐ Sequential ☐ Direct	
Record Size (Characters): 250	Average Number of Records: 20,000
Number of Records: Maximum: 50,000	
Percent Growth per Year: 10%	
Data Set Name: Customer.MST	
Copy Member: Custmast	
Data Structure: Customer Record	
Primary key: Customer Number	
Secondary keys: Customer Name, Telephone	

Comments: Customer Master records are archived to a history file if the customer has not made a purchase within the past two years to optimize database performance.

Table 4.2-4: Data Store Description – Item Master

DATA STORE DESCRIPTION FORM (ID: D2)	
Name: Item Master	
Alias: Product Inventory, SKU File	
Description: Contains a permanent record of all products available for sale, including pricing, descriptions, and stock levels at Circle K stores.	
Data Store Characteristics:	
File Type: ☒ Computer ☐ Manual	
File Format: ☒ Database ☐ Indexed ☐ Sequential ☐ Direct	
Record Size (Characters): 180	Average Number of Records: 5,000
Number of Records: Maximum: 10,000	
Percent Growth per Year: 5%	
Data Set Name: Item.MST	
Copy Member: Itemmast	
Data Structure: Item Record	
Primary key: Item Number	
Secondary keys: Item Description, Category	
Comments: This store is integrated with the real-time inventory system to reflect current availability for the CK GO application.	

4.3 The Importance of Maintaining a Data Dictionary to Prevent Information Discrepancies

As an information system like CK GO scales and multiple team members collaborate on its design, discrepancies often arise when different developers use different aliases, formats, or structural rules for the same data entity. To mitigate this risk and ensure seamless collaboration, maintaining a robust data dictionary is imperative.

According to Kendall and Kendall (2020), "a data dictionary is a reference work of data about data (that is, metadata)". Systems analysts compile these dictionaries to guide them through the entire analysis and design process. It acts as a centralized repository that establishes a single source of truth for the project. As Kendall and Kendall emphasize, "a data dictionary is a document that collects and coordinates specific data terms, and it confirms what each term means to different people in the organization". By strictly defining the inputs, outputs, data structures, and file formats, the analysis team avoids miscommunications and overlapping work.

Furthermore, "one important reason for maintaining a data dictionary is to keep clean data. This means that data must be consistent". For instance, if one developer stores a customer's gender as "M" and another developer on the same team stores it as "Male", the system's data is no longer clean or consistent. A data dictionary solves this by standardizing definitions. In modern systems, automated data dictionaries are especially valuable for their capacity to "cross-reference data items, thereby allowing necessary program changes to all applications that share a common element". This feature prevents developers from updating programs on a haphazard basis, ensuring that a modification in one part of the system is uniformly implemented across all -related applications.

Ultimately, a well-maintained data dictionary ensures that all components of the system are uniformly understood and seamlessly integrated, preventing costly architectural errors and information discrepancies during the development phase.

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